

## Module 1

### Personal Development Workouts

1. Prepare a vision board.
2. Prepare and start practicing your affirmations.

*Write a short description of personal task 1*

*Link to the image of your vision board*

*Write a short description about personal task 2*

*Link to your affirmations*

## Foundation Week

### Technical Workouts

1. Understand Artificial Intelligence & Machine Learning
  - a. What is Artificial Intelligence
  - b. What is Machine Learning
  - c. AI vs ML vs Deep Learning
  - d. Real-world applications of AI systems
  - e. Industry use cases of Machine Learning
2. Understand the Machine Learning Lifecycle
  - a. Data collection
  - b. Data preparation
  - c. Model training
  - d. Model evaluation
  - e. Model deployment
  - f. Continuous improvement of models
3. Understand the Role of Data in ML
  - a. Structured vs unstructured data
  - b. Features and labels
  - c. Training and testing datasets
  - d. Importance of data quality

- e. Data-driven decision-making
- 4. Python Basics
  - a. Variables & Data Types
  - b. Operators
  - c. Control Statements
  - d. Input & Output
- 5. Practice some basic logical problems like prime numbers, unique numbers, etc...
- 6. Basic Statistics(Mean, Median, Mode, Variance, Standard Deviation ,Range & Quartiles)
- 7. Basic Mathematics
  - a. Vectors and Matrices (addition, multiplication)
  - b. Basic Probability (independent & dependent events)
- 8. Understand Modern AI Systems
  - a. Traditional ML systems
  - b. Deep Learning systems
  - c. Generative AI systems
  - d. Large Language Models (LLMs)
  - e. Agentic AI systems
- 9. Understand Development Environment Setup
  - a. Python installation
  - b. Package management basics
  - c. Virtual environments
  - d. Jupyter Notebook setup
  - e. VS Code setup
  - f. AI-assisted development tools
- 10. Understand Problem Solving for ML
  - a. Thinking in inputs and outputs
  - b. Breaking problems into smaller steps
  - c. Understanding patterns in data
  - d. Experimentation mindset
  - e. Observation-driven problem solving
- 11. Understand AI-Assisted Learning
  - a. Using AI for:

- i. Concept explanations
- ii. Code explanations
- iii. Learning assistance
- iv. Research assistance
- b. Risks of blindly trusting AI outputs
- c. Importance of validation
- d. Asking effective technical questions
- e. Building learning workflows using AI

#### 12. Apply Through Hands-on Tasks

- a. Set up ML development environment
- b. Explore sample datasets
- c. Analyze real-world ML applications
- d. Document ML lifecycle for a chosen problem
- e. Compare traditional software vs ML systems
- f. Use AI to explain an ML workflow and validate its explanation

### Miscellaneous Workouts

1. Practice typing for at least one hour each day on Typing Club, <http://typingclub.com/>. Finish as many chapters as possible. Don't spend more than an hour each day.
2. Prepare a topic for the tech seminar. Record and upload it on YouTube as an unlisted video.  
Topic eg: TCP/IP
3. Reflect on your week's learning experience. Were you satisfied with your progress? How do you plan to perform in the upcoming review?  
What new concepts or skills did you learn, and how have they contributed to your growth?  
Reflect on your last review—what mistakes did you make, and how did you prepare to address them? Finally, what strategies or methods did you adopt differently this week?
4. Prepare your progress video for the last week. Record and upload it on YouTube as an unlisted video.

*Write a short description about this task (Miscellaneous Workouts 1)*

*Link to screenshot image*

*Write a short description about this task (Miscellaneous Workouts 2)*

*Link to your seminar video*

*Write an essay of not less than 150 words.*

*Write a short description about this task (Miscellaneous Workouts 3)*

*Link to your progress video*